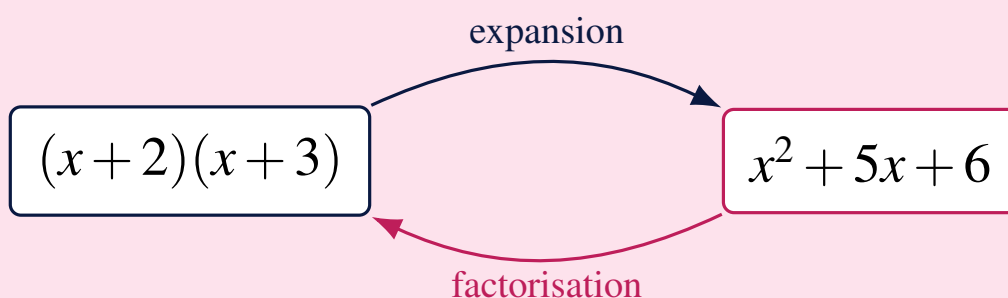


Factorisation

Notes · Practice by Type · Mixed Practice

Common factors · Difference of two squares · Perfect squares · Grouping
Sum & product · Splitting the middle term ($ax^2 + bx + c$)



How to use this booklet

- Read the pink **Key Notes** pages first.
- Sections 1–12 practise one type of factorisation at a time. Section 13 is **mixed practice**: you decide the method.
- Each white **Question** page is followed by a yellow **Answer** page; each answer is in the same position as its question.
- Always check a factorisation by expanding it again.

Key Notes

★ GOLDEN RULE ★

Always take out the common factor FIRST

Then factorise what is left in the bracket.

$$3x^2 - 27 = 3(x^2 - 9) = 3(x + 3)(x - 3)$$

1. What is factorisation?

Factorisation writes an expression as a **product of factors**. It is the reverse of expanding.

$$x^2 + 5x + 6 = (x + 2)(x + 3) \quad (x + 2) \text{ and } (x + 3) \text{ are factors.}$$

Key identities $(x + p)(x + q) = x^2 + (p + q)x + pq$ $(x \pm a)^2 = x^2 \pm 2ax + a^2$
 $(x + a)(x - a) = x^2 - a^2$

2. Common factors — always do this first

Take out the **HCF** (highest common factor) of all the terms.

$$2x^2 + 3x = x(2x + 3) \quad -2x^2 - 6x = -2x(x + 3) \text{ (negative HCF)}$$

A **bracket** can be the common factor:

$$(x - 5)^2 - 2(x - 5) = (x - 5)[(x - 5) - 2] = (x - 5)(x - 7)$$
$$(x + 2)^2 + 2x + 4 = (x + 2)^2 + 2(x + 2) = (x + 2)(x + 4)$$

3. Difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

$$4x^2 - 25 = (2x)^2 - 5^2 = (2x + 5)(2x - 5)$$

$$2x^2 - 8 = 2(x^2 - 4) = 2(x + 2)(x - 2) \quad (\text{common factor first})$$

$$x^2 - 7 = (x + \sqrt{7})(x - \sqrt{7}) \quad (\text{surds})$$

$$(x + 2)^2 - (x - 1)^2 = [(x + 2) + (x - 1)][(x + 2) - (x - 1)] = 3(2x + 1)$$

A **sum** of two squares, e.g. $x^2 + 4$, does **not** factorise.

4. Perfect squares

$$x^2 + 2ax + a^2 = (x + a)^2 \quad x^2 - 2ax + a^2 = (x - a)^2$$

Check: the first and last terms are squares, and the middle term is $2 \times \sqrt{\text{first}} \times \sqrt{\text{last}}$.

$$x^2 - 14x + 49 = x^2 - 2(7)x + 7^2 = (x - 7)^2$$

$$9x^2 - 6x + 1 = (3x)^2 - 2(3x)(1) + 1^2 = (3x - 1)^2$$

$$-8x^2 - 24x - 18 = -2(4x^2 + 12x + 9) = -2(2x + 3)^2$$

5. Four terms — grouping in pairs

Group into two pairs, factorise each pair, then take out the common bracket.

$$ax + by + bx + ay = a(x + y) + b(x + y) = (x + y)(a + b)$$

$$2x^2 - 15 + 3x - 10x = 2x(x - 5) + 3(x - 5) = (x - 5)(2x + 3)$$

$$x^2 + 4x + 4 - m^2 = (x + 2)^2 - m^2 = (x + 2 + m)(x + 2 - m) \quad (\text{group 3 terms} + 1)$$

6. Trinomials $x^2 + bx + c$ — sum and product

Find two numbers p and q with **sum** $p + q = b$ and **product** $pq = c$. Then $x^2 + bx + c = (x + p)(x + q)$.

$$x^2 + 5x + 4 : \text{sum } 5, \text{ product } 4 \Rightarrow 1, 4 \quad = (x + 1)(x + 4)$$

$$x^2 - x - 12 : \text{sum } -1, \text{ product } -12 \Rightarrow -4, 3 \quad = (x - 4)(x + 3)$$

$$3x^2 - 9x + 6 = 3(x^2 - 3x + 2) \quad = 3(x - 2)(x - 1)$$

Signs: if $c > 0$ both numbers have the sign of b ; if $c < 0$ one is positive and one negative.

7. Trinomials $ax^2 + bx + c$ ($a \neq 1$) — splitting the middle term

1. Work out ac . Find two numbers with **product** ac and **sum** b .
2. Split bx into these two terms.
3. Group in pairs and factorise.

$$6x^2 + 19x + 10 : ac = 60, \text{ numbers } 4 \text{ and } 15$$

$$= 6x^2 + 4x + 15x + 10 = 2x(3x + 2) + 5(3x + 2) = (3x + 2)(2x + 5)$$

$$-5x^2 - 7x + 6 = -(5x^2 + 7x - 6) : ac = -30, \text{ numbers } 10 \text{ and } -3$$

$$= -(5x^2 + 10x - 3x - 6) = -(x + 2)(5x - 3)$$

8. Which method? — checklist

1. Take out any **common factor** (including -1).
2. Two terms? Look for a **difference of two squares**.
3. Three terms? Look for a **perfect square**, else **sum & product** ($a = 1$) or **split the middle term** ($a \neq 1$).
4. Four terms? **Group** in pairs.
5. Check each bracket: can it be factorised again? Then **expand to check**.

Fully factorise by first removing a common factor.

★ **Golden rule: always take out the common factor first!**

1. $3x^2 + 5x$

2. $2x^2 - 7x$

3. $3x^2 + 6x$

4. $4x^2 - 8x$

5. $-2x^2 + 9x$

6. $-3x^2 - 15x$

7. $-4x + 8x^2$

8. $-5x - 10x^2$

9. $12x - 4x^2$

1. $x(3x + 5)$

2. $x(2x - 7)$

3. $3x(x + 2)$

4. $4x(x - 2)$

5. $-x(2x - 9)$

6. $-3x(x + 5)$

7. $4x(2x - 1)$

8. $-5x(1 + 2x)$

9. $4x(3 - x)$

Fully factorise by first removing a common factor.

★ **Golden rule: always take out the common factor first!**

10. $x^3 + x^2 + x$

11. $2x^3 + 11x^2 + 4x$

12. $ab + ac + ad$

13. $ax^2 + 2ax$

14. $a^2b + ab^2$

15. $ax^3 + ax^2$

16. $3x^2 + 12x$

17. $15x - 3x^2$

18. $16x - 8x^2$

10. $x(x^2 + x + 1)$

11. $x(2x^2 + 11x + 4)$

12. $a(b + c + d)$

13. $ax(x + 2)$

14. $ab(a + b)$

15. $ax^2(x + 1)$

16. $3x(x + 4)$

17. $3x(5 - x)$

18. $8x(2 - x)$

Fully factorise by removing a common (bracket) factor.

★ **Golden rule: always take out the common factor first!**

19. $(x + 2)^2 - 5(x + 2)$

20. $(x - 1)^2 - 3(x - 1)$

21. $(x + 1)^2 + 2(x + 1)$

22. $(x - 2)^2 + 3x - 6$

23. $x + 3 + (x + 3)^2$

24. $(x + 4)^2 + 8 + 2x$

25. $(x - 3)^2 - x + 3$

26. $(x + 4)^2 - 2x - 8$

27. $(x - 4)^2 - 5x + 20$

19. $(x + 2)(x - 3)$

20. $(x - 1)(x - 4)$

21. $(x + 1)(x + 3)$

22. $(x - 2)(x + 1)$

23. $(x + 3)(x + 4)$

24. $(x + 4)(x + 6)$

25. $(x - 3)(x - 4)$

26. $(x + 4)(x + 2)$

27. $(x - 4)(x - 9)$

Fully factorise by removing a common (bracket) factor.

★ **Golden rule: always take out the common factor first!**

28. $3x + 6 + (x + 2)^2$

29. $(x + 1)^3 + (x + 1)^2$

30. $(a + b)^3 + a + b$

31. $2(x + 1)^2 + x + 1$

32. $3(x - 2)^2 - (x - 2)$

33. $4(a + b)^2 - 2a - 2b$

34. $(x + 3)^2 - 4(x + 3)$

35. $(2x - 1)^2 + 2x - 1$

28. $(x + 2)(x + 5)$

29. $(x + 1)^2(x + 2)$

30. $(a + b)((a + b)^2 + 1)$

31. $(x + 1)(2x + 3)$

32. $(x - 2)(3x - 7)$

33. $2(a + b)(2a + 2b - 1)$

34. $(x + 3)(x - 1)$

35. $2x(2x - 1)$

Use $a^2 - b^2 = (a + b)(a - b)$ to fully factorise.

★ Golden rule: always take out the common factor first!

36. $x^2 - 4$

37. $4 - x^2$

38. $x^2 - 81$

39. $25 - x^2$

40. $4x^2 - 1$

41. $9x^2 - 16$

42. $4x^2 - 9$

43. $36 - 49x^2$

44. $3x^2 - 27$

36. $(x+2)(x-2)$

37. $(2+x)(2-x)$

38. $(x+9)(x-9)$

39. $(5+x)(5-x)$

40. $(2x+1)(2x-1)$

41. $(3x+4)(3x-4)$

42. $(2x+3)(2x-3)$

43. $(6+7x)(6-7x)$

44. $3(x+3)(x-3)$

Use $a^2 - b^2 = (a + b)(a - b)$ to fully factorise.

★ Golden rule: always take out the common factor first!

45. $-2x^2 + 8$

46. $3x^2 - 75$

47. $-5x^2 + 5$

48. $8x^2 - 18$

49. $-27x^2 + 75$

50. $6x^2 - 24y^2$

51. $9 - 25x^2$

52. $18 - 2a^2$

45. $-2(x+2)(x-2)$

46. $3(x+5)(x-5)$

47. $-5(x+1)(x-1)$

48. $2(2x+3)(2x-3)$

49. $-3(3x+5)(3x-5)$

50. $6(x+2y)(x-2y)$

51. $(3+5x)(3-5x)$

52. $2(3+a)(3-a)$

If possible, factorise into linear factors.

★ **Golden rule: always take out the common factor first!**

53. $x^2 - 3$

54. $x^2 + 4$

55. $x^2 - 15$

56. $3x^2 - 15$

57. $(x + 1)^2 - 6$

58. $(x + 2)^2 + 6$

59. $(x - 2)^2 - 7$

60. $(x + 3)^2 - 17$

61. $(x - 4)^2 + 9$

62. $x^2 - 13$

63. $x^2 - 23$

53. $(x + \sqrt{3})(x - \sqrt{3})$

54. not factorisable

55. $(x + \sqrt{15})(x - \sqrt{15})$

56. $3(x + \sqrt{5})(x - \sqrt{5})$

57. $(x + 1 + \sqrt{6})(x + 1 - \sqrt{6})$

58. not factorisable

59. $(x - 2 + \sqrt{7})(x - 2 - \sqrt{7})$

60. $(x + 3 + \sqrt{17})(x + 3 - \sqrt{17})$

61. not factorisable

62. $(x + \sqrt{13})(x - \sqrt{13})$

63. $(x + \sqrt{23})(x - \sqrt{23})$

If possible, factorise into linear factors.

★ **Golden rule: always take out the common factor first!**

64. $(x + 2)^2 - 3$

65. $(x + 1)^2 - 4$

66. $(2x + 1)^2 - 9$

67. $(1 - x)^2 - 16$

68. $(x + 3)^2 - 4x^2$

69. $4x^2 - (x + 2)^2$

70. $9x^2 - (3 - x)^2$

71. $(2x + 1)^2 - (x - 2)^2$

72. $(3x - 1)^2 - (x + 1)^2$

73. $4x^2 - (2x + 3)^2$

64. $(x + 2 + \sqrt{3})(x + 2 - \sqrt{3})$

65. $(x + 3)(x - 1)$

66. $(2x + 4)(2x - 2)$

$= 4(x + 2)(x - 1)$

67. $(5 - x)(-3 - x)$

$= (x - 5)(x + 3)$

68. $(3x + 3)(3 - x)$

$= 3(x + 1)(3 - x)$

69. $(3x + 2)(x - 2)$

70. $(2x + 3)(4x - 3)$

71. $(3x - 1)(x + 3)$

72. $(4x)(2x - 2)$

$= 8x(x - 1)$

73. $-3(4x + 3)$

Use the perfect square rules to fully factorise.

★ Golden rule: always take out the common factor first!

74. $x^2 + 6x + 9$

75. $x^2 + 8x + 16$

76. $x^2 - 6x + 9$

77. $x^2 - 8x + 16$

78. $x^2 + 2x + 1$

79. $x^2 - 10x + 25$

80. $y^2 + 18y + 81$

81. $m^2 - 20m + 100$

82. $t^2 + 12t + 36$

83. $9x^2 + 6x + 1$

74. $(x + 3)^2$

75. $(x + 4)^2$

76. $(x - 3)^2$

77. $(x - 4)^2$

78. $(x + 1)^2$

79. $(x - 5)^2$

80. $(y + 9)^2$

81. $(m - 10)^2$

82. $(t + 6)^2$

83. $(3x + 1)^2$

Use the perfect square rules to fully factorise.

★ Golden rule: always take out the common factor first!

84. $4x^2 - 4x + 1$

85. $9x^2 + 12x + 4$

86. $25x^2 - 10x + 1$

87. $16x^2 + 24x + 9$

88. $25x^2 - 20x + 4$

89. $-x^2 + 2x - 1$

90. $-2x^2 - 8x - 8$

91. $-3x^2 - 30x - 75$

92. $x^2 - 12x + 36$

93. $2x^2 + 8x + 8$

84. $(2x - 1)^2$

85. $(3x + 2)^2$

86. $(5x - 1)^2$

87. $(4x + 3)^2$

88. $(5x - 2)^2$

89. $-(x - 1)^2$

90. $-2(x + 2)^2$

91. $-3(x + 5)^2$

92. $(x - 6)^2$

93. $2(x + 2)^2$

Fully factorise by grouping the terms in pairs.

★ **Golden rule: always take out the common factor first!**

94. $bx + cx + by + cy$

95. $2px + 3q + 2qx + 3p$

96. $6ax + 3bx + 2b + 4a$

97. $am - bn - an + bm$

98. $3dr + r - 3ds - s$

99. $2ac - 5a + 2bc - 5b$

100. $x^2 + 5x + 7x + 35$

101. $x^2 - 2x - 6x + 12$

102. $x^2 + 3x + 9 + 3x$

94. $(x + y)(b + c)$

95. $(p + q)(2x + 3)$

96. $(2a + b)(3x + 2)$

97. $(a + b)(m - n)$

98. $(3d + 1)(r - s)$

99. $(a + b)(2c - 5)$

100. $(x + 5)(x + 7)$

101. $(x - 2)(x - 6)$

102. $(x + 3)^2$

Fully factorise by grouping the terms in pairs.

★ **Golden rule: always take out the common factor first!**

103. $x^2 + 8 + 8x + x$

104. $2x^2 + 3x + 3 + 2x$

105. $3x^2 + x - 3x - 1$

106. $2x^2 + 3x + 10x + 15$

107. $6x^2 - 3x - 2 + 4x$

108. $4x^2 + x + 8x + 2$

109. $6x^2 + 3x + 10x + 5$

110. $6x^2 - 4x + 9x - 6$

111. $4x^2 + x - 8x - 2$

103. $(x + 1)(x + 8)$

104. $(x + 1)(2x + 3)$

105. $(3x + 1)(x - 1)$

106. $(2x + 3)(x + 5)$

107. $(2x - 1)(3x + 2)$

108. $(4x + 1)(x + 2)$

109. $(2x + 1)(3x + 5)$

110. $(3x - 2)(2x + 3)$

111. $(4x + 1)(x - 2)$

Fully factorise by grouping the terms in pairs.

★ **Golden rule: always take out the common factor first!**

112. $3x^2 + 4x + 33x + 44$

113. $18x^2 + 3x - 12x - 2$

114. $10x^2 + 4x - 35x - 14$

115. $ax + 2a + 2b + bx$

116. $2cx - 2dx + d - c$

117. $mx + nx - my - ny$

118. $3a^2 + ab - 2b^2 - 6ab$

112. $(3x + 4)(x + 11)$

113. $(6x + 1)(3x - 2)$

114. $(5x + 2)(2x - 7)$

115. $(x + 2)(a + b)$

116. $(c - d)(2x - 1)$

117. $(m + n)(x - y)$

118. $(a - 2b)(3a + b)$

Fully factorise.

★ **Golden rule: always take out the common factor first!**

119. $x^2 - 2x + 1 - a^2$

120. $x^2 - a^2 + x + a$

121. $b^2 - x^2 - 4x - 4$

122. $c^2 - x^2 + 6x - 9$

123. $x^2 - y^2 + y - x$

124. $x^2 + 4x + 4 - m^2$

125. $x^2 + 2ax + a^2 - b^2$

126. $x^2 - y^2 - 3x - 3y$

119. $(x - 1 + a)(x - 1 - a)$

120. $(x + a)(x - a + 1)$

121. $(b + x + 2)(b - x - 2)$

122. $(c + x - 3)(c - x + 3)$

123. $(x - y)(x + y - 1)$

124. $(x + 2 + m)(x + 2 - m)$

125. $(x + a + b)(x + a - b)$

126. $(x + y)(x - y - 3)$

Use the sum and product method: find p and q with $p + q = b$ and $pq = c$.

★ Golden rule: always take out the common factor first!

127. $x^2 + 3x + 2$

128. $x^2 + 5x + 6$

129. $x^2 - x - 6$

130. $x^2 + 3x - 10$

131. $x^2 + 4x - 21$

132. $x^2 - 14x + 49$

133. $x^2 + 3x - 28$

134. $x^2 + 7x + 10$

135. $x^2 - 11x + 24$

136. $x^2 + 15x + 44$

137. $x^2 + x - 42$

138. $x^2 - x - 56$

139. $x^2 - 18x + 81$

140. $x^2 - 4x - 32$

127. $(x + 1)(x + 2)$

numbers 1 and 2

128. $(x + 2)(x + 3)$

numbers 2 and 3

129. $(x - 3)(x + 2)$

numbers -3 and 2

130. $(x + 5)(x - 2)$

numbers -2 and 5

131. $(x + 7)(x - 3)$

numbers -3 and 7

132. $(x - 7)^2$

numbers -7 and -7

133. $(x + 7)(x - 4)$

numbers -4 and 7

134. $(x + 2)(x + 5)$

numbers 2 and 5

135. $(x - 3)(x - 8)$

numbers -8 and -3

136. $(x + 4)(x + 11)$

numbers 4 and 11

137. $(x + 7)(x - 6)$

numbers -6 and 7

138. $(x - 8)(x + 7)$

numbers -8 and 7

139. $(x - 9)^2$

numbers -9 and -9

140. $(x - 8)(x + 4)$

numbers -8 and 4

Fully factorise by first removing a common factor.

★ **Golden rule: always take out the common factor first!**

141. $2x^2 - 6x - 8$

142. $3x^2 + 9x - 12$

143. $5x^2 + 10x - 15$

144. $4x^2 + 4x - 80$

145. $2x^2 - 4x - 30$

146. $3x^2 + 12x - 63$

147. $-2x^2 + 2x + 40$

148. $-3x^2 + 12x - 12$

149. $-7x^2 - 21x + 28$

150. $-x^2 - 3x - 2$

141. $2(x-4)(x+1)$

HCF 2; numbers -4 and 1

142. $3(x+4)(x-1)$

HCF 3; numbers -1 and 4

143. $5(x+3)(x-1)$

HCF 5; numbers -1 and 3

144. $4(x+5)(x-4)$

HCF 4; numbers -4 and 5

145. $2(x-5)(x+3)$

HCF 2; numbers -5 and 3

146. $3(x+7)(x-3)$

HCF 3; numbers -3 and 7

147. $-2(x-5)(x+4)$

HCF -2 ; numbers -5 and 4

148. $-3(x-2)^2$

HCF -3 ; numbers -2 and -2

149. $-7(x+4)(x-1)$

HCF -7 ; numbers -1 and 4

150. $-(x+1)(x+2)$

HCF -1 ; numbers 1 and 2

Fully factorise by first removing a common factor.

★ **Golden rule: always take out the common factor first!**

151. $-x^2 + 5x - 6$

152. $-x^2 + 9x - 18$

153. $5x^2 + 15x - 50$

154. $-2x^2 - 8x + 42$

155. $4x - x^2 + 32$

156. $2x^2 + 8x + 6$

157. $5x^2 - 10x + 5$

158. $3x^2 - 6x - 9$

159. $7x^2 + 28x + 28$

151. $-(x-2)(x-3)$

HCF -1 ; numbers -3 and -2

152. $-(x-3)(x-6)$

HCF -1 ; numbers -6 and -3

153. $5(x+5)(x-2)$

HCF 5 ; numbers -2 and 5

154. $-2(x+7)(x-3)$

HCF -2 ; numbers -3 and 7

155. $-(x-8)(x+4)$

HCF -1 ; numbers -8 and 4

156. $2(x+1)(x+3)$

HCF 2 ; numbers 1 and 3

157. $5(x-1)^2$

HCF 5 ; numbers -1 and -1

158. $3(x-3)(x+1)$

HCF 3 ; numbers -3 and 1

159. $7(x+2)^2$

HCF 7 ; numbers 2 and 2

Fully factorise by splitting the middle term: find two numbers with product ac and sum b .

★ Golden rule: always take out the common factor first!

160. $2x^2 + 5x + 3$

161. $2x^2 + 7x + 5$

162. $7x^2 + 9x + 2$

163. $3x^2 + 7x + 4$

164. $3x^2 + 13x + 4$

165. $3x^2 + 8x + 4$

166. $8x^2 + 14x + 3$

167. $21x^2 + 17x + 2$

160. $(2x + 3)(x + 1)$

$ac = 6$: 2 and 3

161. $(2x + 5)(x + 1)$

$ac = 10$: 2 and 5

162. $(7x + 2)(x + 1)$

$ac = 14$: 2 and 7

163. $(3x + 4)(x + 1)$

$ac = 12$: 3 and 4

164. $(3x + 1)(x + 4)$

$ac = 12$: 1 and 12

165. $(3x + 2)(x + 2)$

$ac = 12$: 2 and 6

166. $(4x + 1)(2x + 3)$

$ac = 24$: 2 and 12

167. $(7x + 1)(3x + 2)$

$ac = 42$: 3 and 14

Fully factorise by splitting the middle term: find two numbers with product ac and sum b .

★ Golden rule: always take out the common factor first!

168. $6x^2 + 5x + 1$

169. $6x^2 + 19x + 3$

170. $10x^2 + 17x + 3$

171. $14x^2 + 37x + 5$

172. $3x^2 + 14x + 8$

173. $12x^2 + 17x + 6$

174. $12x^2 + 7x + 1$

175. $2x^2 + 17x + 8$

168. $(3x + 1)(2x + 1)$

$ac = 6$: 2 and 3

169. $(6x + 1)(x + 3)$

$ac = 18$: 1 and 18

170. $(5x + 1)(2x + 3)$

$ac = 30$: 2 and 15

171. $(7x + 1)(2x + 5)$

$ac = 70$: 2 and 35

172. $(3x + 2)(x + 4)$

$ac = 24$: 2 and 12

173. $(4x + 3)(3x + 2)$

$ac = 72$: 8 and 9

174. $(4x + 1)(3x + 1)$

$ac = 12$: 3 and 4

175. $(2x + 1)(x + 8)$

$ac = 16$: 1 and 16

Fully factorise.

★ **Golden rule: always take out the common factor first!**

176. $2x^2 - 9x - 5$

177. $3x^2 + 5x - 2$

178. $3x^2 - 5x - 2$

179. $2x^2 + 3x - 2$

180. $2x^2 + 3x - 5$

181. $5x^2 - 14x - 3$

182. $5x^2 - 8x + 3$

183. $11x^2 - 9x - 2$

184. $3x^2 - 7x - 6$

185. $2x^2 - 3x - 9$

186. $3x^2 - 17x + 10$

176. $(2x + 1)(x - 5)$

$ac = -10$: -10 and 1

177. $(3x - 1)(x + 2)$

$ac = -6$: -1 and 6

178. $(3x + 1)(x - 2)$

$ac = -6$: -6 and 1

179. $(2x - 1)(x + 2)$

$ac = -4$: -1 and 4

180. $(2x + 5)(x - 1)$

$ac = -10$: -2 and 5

181. $(5x + 1)(x - 3)$

$ac = -15$: -15 and 1

182. $(5x - 3)(x - 1)$

$ac = 15$: -5 and -3

183. $(11x + 2)(x - 1)$

$ac = -22$: -11 and 2

184. $(3x + 2)(x - 3)$

$ac = -18$: -9 and 2

185. $(2x + 3)(x - 3)$

$ac = -18$: -6 and 3

186. $(3x - 2)(x - 5)$

$ac = 30$: -15 and -2

Fully factorise.

★ **Golden rule: always take out the common factor first!**

187. $5x^2 - 13x - 6$

188. $3x^2 + 10x - 8$

189. $2x^2 + 17x - 9$

190. $2x^2 + 9x - 18$

191. $2x^2 + 11x - 21$

192. $15x^2 + x - 2$

193. $21x^2 - 62x - 3$

194. $9x^2 - 12x + 4$

195. $12x^2 + 17x - 40$

196. $16x^2 + 34x - 15$

197. $15x^2 + 14x - 8$

187. $(5x + 2)(x - 3)$

$ac = -30$: -15 and 2

188. $(3x - 2)(x + 4)$

$ac = -24$: -2 and 12

189. $(2x - 1)(x + 9)$

$ac = -18$: -1 and 18

190. $(2x - 3)(x + 6)$

$ac = -36$: -3 and 12

191. $(2x - 3)(x + 7)$

$ac = -42$: -3 and 14

192. $(5x + 2)(3x - 1)$

$ac = -30$: -5 and 6

193. $(21x + 1)(x - 3)$

$ac = -63$: -63 and 1

194. $(3x - 2)^2$

$ac = 36$: -6 and -6

195. $(4x - 5)(3x + 8)$

$ac = -480$: -15 and 32

196. $(8x - 3)(2x + 5)$

$ac = -240$: -6 and 40

197. $(5x - 2)(3x + 4)$

$ac = -120$: -6 and 20

Fully factorise by first removing -1 as a common factor.

★ **Golden rule: always take out the common factor first!**

198. $-3x^2 - x + 14$

199. $-5x^2 + 11x - 2$

200. $-4x^2 - 9x + 9$

201. $-9x^2 + 12x - 4$

202. $-8x^2 - 14x - 3$

203. $-12x^2 + 16x + 3$

198. $-(3x + 7)(x - 2)$

take out -1 ; $ac = -42$: -6 and 7

199. $-(5x - 1)(x - 2)$

take out -1 ; $ac = 10$: -10 and -1

200. $-(4x - 3)(x + 3)$

take out -1 ; $ac = -36$: -3 and 12

201. $-(3x - 2)^2$

take out -1 ; $ac = 36$: -6 and -6

202. $-(4x + 1)(2x + 3)$

take out -1 ; $ac = 24$: 2 and 12

203. $-(6x + 1)(2x - 3)$

take out -1 ; $ac = -36$: -18 and 2

Fully factorise where possible. Decide which method to use for each one.

★ **Golden rule: always take out the common factor first!**

204. $3x^2 + 9x$

205. $5x^2 - 15$

206. $3x - 5x^2$

207. $x^2 + 3x - 40$

208. $2x^2 - 32$

209. $x^2 + 9$

210. $x^2 + 10x + 25$

211. $x^2 - 16x + 39$

212. $x^2 - 7x - 60$

213. $x^2 - 2x - 8$

214. $x^2 + 11x + 30$

215. $x^2 + 6x - 16$

204. $3x(x+3)$

205. $5(x^2 - 3)$
 $= 5(x + \sqrt{3})(x - \sqrt{3})$

206. $x(3 - 5x)$

207. $(x+8)(x-5)$

208. $2(x+4)(x-4)$

209. not factorisable

210. $(x+5)^2$

211. $(x-3)(x-13)$

212. $(x-12)(x+5)$

213. $(x-4)(x+2)$

214. $(x+5)(x+6)$

215. $(x+8)(x-2)$

Fully factorise where possible. Decide which method to use for each one.

★ **Golden rule: always take out the common factor first!**

216. $x^2 - 5x - 24$

217. $3x^2 + 6x - 72$

218. $4x^2 - 8x - 60$

219. $3x^2 - 42x + 99$

220. $-x^2 + 9x - 14$

221. $-x^2 - 13x - 36$

222. $-2x^2 - 14x + 36$

223. $6x^2 + 9x - 2x - 3$

224. $x^2 - 8x + 24 - 3x$

225. $3x^2 + 2x + 8 + 12x$

226. $2x^2 + 15x - 8$

227. $2x^2 - 17x + 8$

216. $(x - 8)(x + 3)$

217. $3(x + 6)(x - 4)$

218. $4(x - 5)(x + 3)$

219. $3(x - 3)(x - 11)$

220. $-(x - 2)(x - 7)$

221. $-(x + 4)(x + 9)$

222. $-2(x + 9)(x - 2)$

223. $(2x + 3)(3x - 1)$

224. $(x - 8)(x - 3)$

225. $(3x + 2)(x + 4)$

226. $(2x - 1)(x + 8)$

227. $(2x - 1)(x - 8)$

Fully factorise where possible. Decide which method to use for each one.

★ **Golden rule: always take out the common factor first!**

228. $6x^2 - 11x - 10$

229. $12x^2 + 5x - 2$

230. $12x^2 - 8x - 15$

231. $3x^2 - 17x - 6$

232. $3x^2 - 19x + 6$

233. $3x^2 + 17x - 6$

234. $12x^2 - 23x - 2$

235. $x^3 - 9x$

236. $2x^3 - 12x^2 + 18x$

237. $x^4 - 16$

238. $5x^2 - 45y^2$

239. $a^2 - b^2 + 2a + 2b$

228. $(3x + 2)(2x - 5)$

229. $(4x - 1)(3x + 2)$

230. $(6x + 5)(2x - 3)$

231. $(3x + 1)(x - 6)$

232. $(3x - 1)(x - 6)$

233. $(3x - 1)(x + 6)$

234. $(12x + 1)(x - 2)$

235. $x(x + 3)(x - 3)$

236. $2x(x - 3)^2$

237. $(x^2 + 4)(x + 2)(x - 2)$

238. $5(x + 3y)(x - 3y)$

239. $(a + b)(a - b + 2)$

Fully factorise where possible. Decide which method to use for each one.

★ **Golden rule: always take out the common factor first!**

240. $4x^2 + 20x + 25$

241. $6x^2 + x - 12$

242. $-6x^2 + 7x + 3$

243. $(x - 3)^2 - 16$

244. $x^2y - xy^2$

245. $x^2 + xy - 6y^2$

246. $8x^2 - 2x - 15$

247. $x^4 - 5x^2 + 4$

248. $2x^2 - 50$

249. $100x^2 - 81$

250. $x^2 - 0.25$

240. $(2x + 5)^2$

241. $(3x - 4)(2x + 3)$

242. $-(3x + 1)(2x - 3)$

243. $(x + 1)(x - 7)$

244. $xy(x - y)$

245. $(x + 3y)(x - 2y)$

246. $(4x + 5)(2x - 3)$

247. $(x - 1)(x + 1)(x - 2)(x + 2)$

248. $2(x + 5)(x - 5)$

249. $(10x + 9)(10x - 9)$

250. $(x + 0.5)(x - 0.5)$